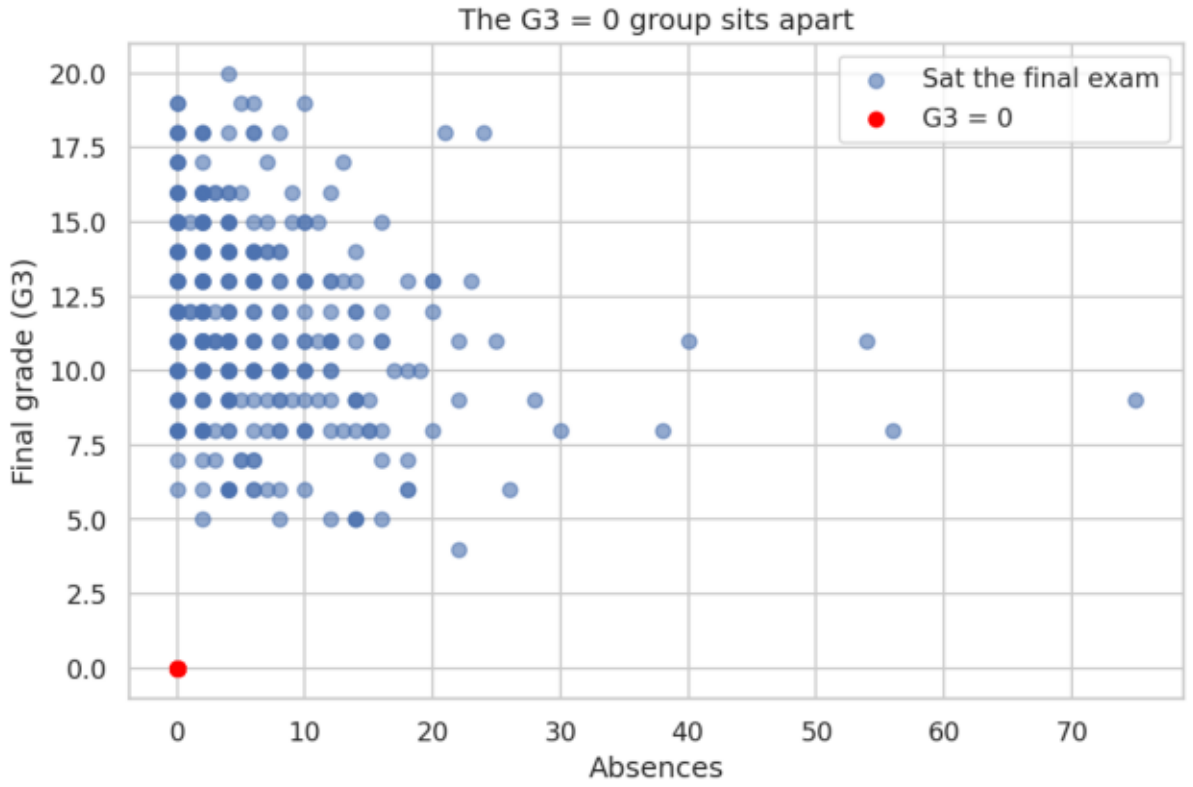
1. The question

- Client: a secondary school's academic office.
- Need: know which students need maths support before the final exam.
- Target: G3, final maths grade (0–20, pass = 10).
- Metric: MAE in grade points, easy for teachers, compared with a 'predict the average' baseline.

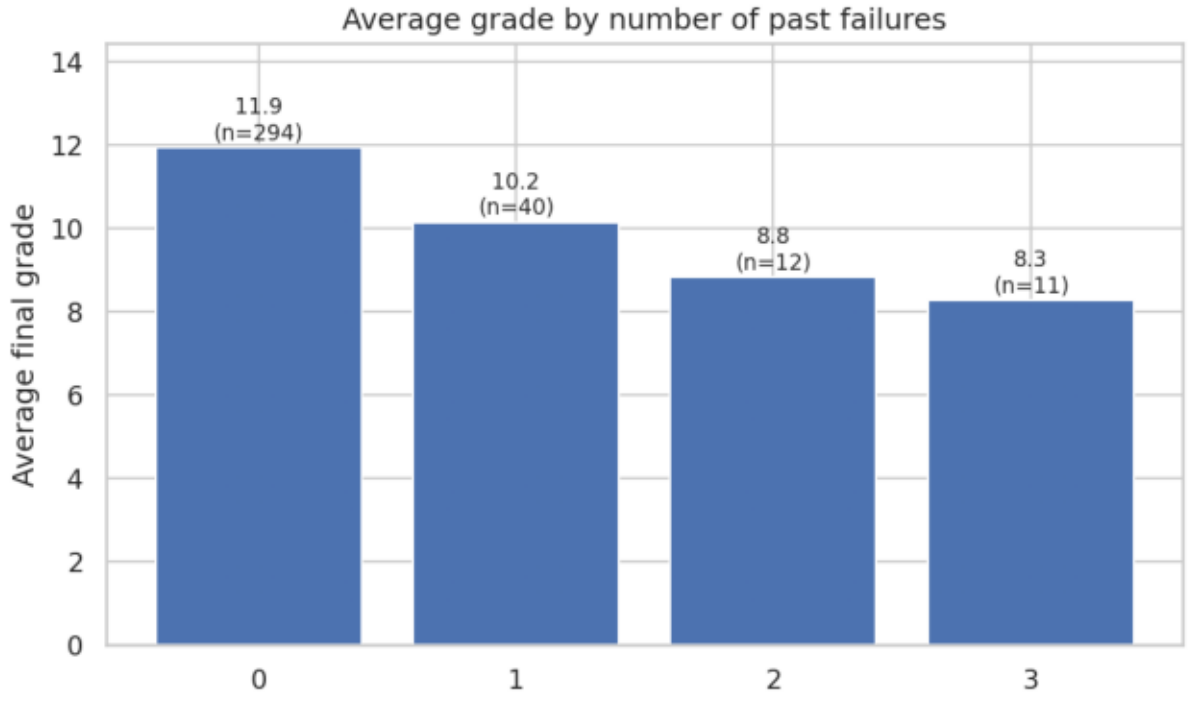
2. The data

- UCI Student Performance, Mathematics file: 395 students × 33 columns.
- Biggest problem: 38 students have G3 = 0 AND 0 absences.
- They did not sit the exam; their 0 hid the real absence effect.
- Fix: flag them (not delete), analyse grades on the 357 who sat, report the 38 separately.



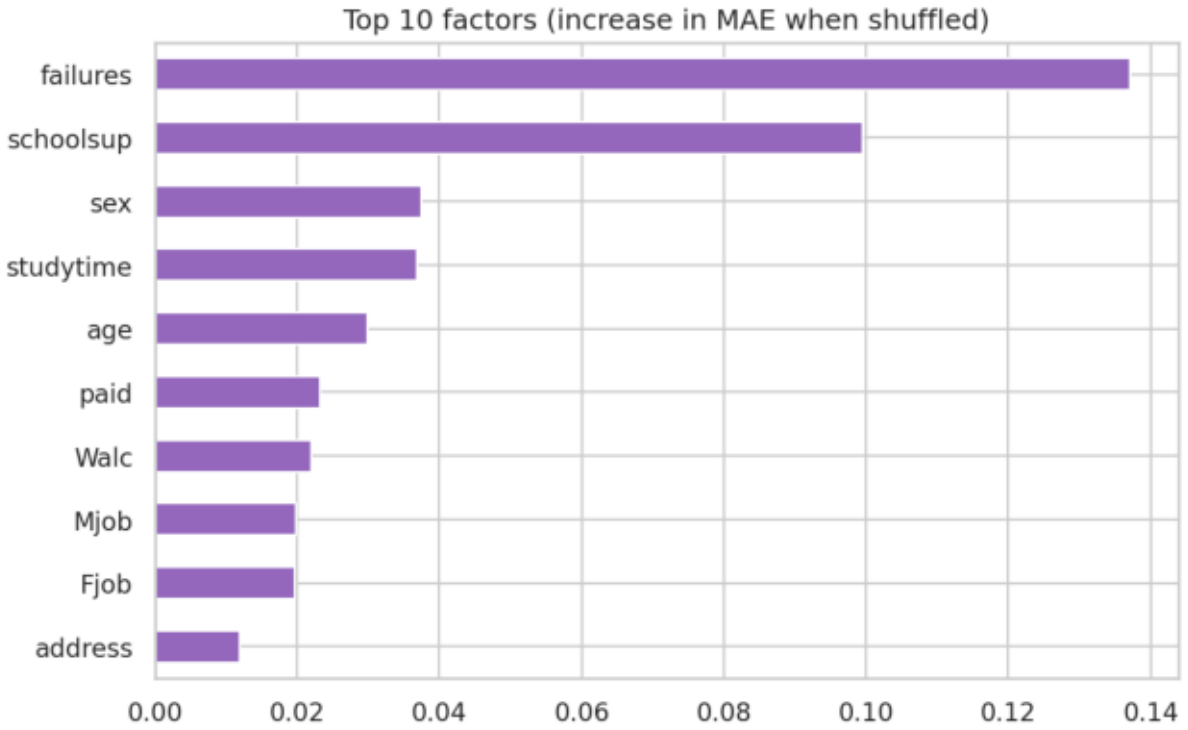
3. Key insights

- Past failures: 11.9 → 8.3 average grade; 3–5× more likely to skip the final.
- Absences matter: 12.6 (none) vs 10.2 (11+).
- School support goes to weaker students, who improve more.



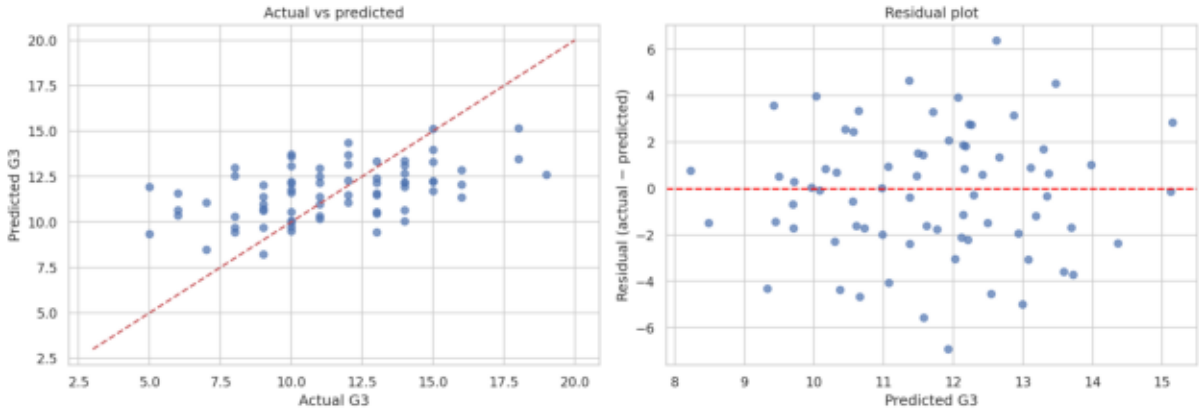
4. The model

- Baseline: predict the average grade → MAE 2.57.
- Final: Random Forest on 30 start-of-year features, random_state=42.
- G1/G2 excluded: they would leak the answer for an early-warning tool.
- Why Random Forest: handles mixed and non-linear features with little tuning.



5. Results

- MAE 2.19 grade points (15% better than baseline); CV 2.34 vs 2.63.
- R^2 0.24: background explains a quarter of grade differences.
- Predictions pulled to the average: failing students predicted ~3 points too high.



6. Recommendation

- Support students with any past failure from week one.
- Attendance alert at ~10 absences; follow up anyone who stops coming.
- Flag students predicted below ~12, not 10.
- Limits: 2 Portuguese schools, self-reported data, R^2 0.24, associations not causes.